

Tracing Recursive Functions

COMS 261

Trace the following recursive functions by making a table with columns for each variable and also columns for the returned expression and its final value. Be sure to indicate what gets printed at the end.

1.

```
1 def reverse(s):  
2     """Reverses a string."""  
3     if len(s) <= 1:  
4         return s  
5     return reverse(s[1:]) + s[0]  
6  
7 print(reverse("reward"))
```

2.

```
1 def gcd(a, b):  
2     """Computes the greatest common divisor of two integers."""  
3     if b == 0:  
4         return a  
5     return gcd(b, a % b)  
6  
7 print(gcd(15, 36))
```

3.

```
1 def pell(n):  
2     """Computes the n-th Pell number."""  
3     if n <= 1:  
4         return n  
5     return 2 * pell(n - 1) + pell(n - 2)  
6  
7 print(pell(5))
```